

RUI DU

Tel.: - ◇ Email: dr397379567@gmail.com ◇ Blog: passerby.github.io

Education

Stevens Institute of Technology
Master in Computer Science

Sep 2025 – Dec 2026
New Jersey, US

Wuchang Shouyi University
Bachelor in Computer Science

Sep 2009 – June 2013
Wuhan, China

Professional Experience

Shanghai Bilibili Technology Co., Ltd.

2016.9 – 2025.4

Senior Machine Learning Researcher, AI Platform Department

- Led RL systems R&D; built and ran a distributed training stack that powered game interaction and platform services; team earned top rankings in IJCAI and NeurIPS challenges.
- Engineered a ZeroMQ/Arrow-based RL platform that decoupled trainer, inference (CPU/GPU), evaluator, monitor, model pool, and data buffer; added population methods (self-play, league, PBT) with real-time Elo/TrueSkill and auto rollback or early stop on regressions.
- Co-designed the RLHF pipeline on A800 clusters; trained a 70B reward model and 16B/1.9B actorcritic; shipped PPO reinforcement of SFT with safety scoring for downstream projects.
- Partnered with game studios to deploy Unity ML-Agents; scaled headless Docker training on Linux clusters; delivered live co-play demos where viewers issued text commands to agents.
- Built the data platform: Hadoop/Hive and Airflow for ETL, SLAs, and governance; KafkaGo streaming cleanseRedis cachehourly persistence to HDFS; Grafana/Rails dashboards with second-level A/B monitoring and alerts for online experiments.

Shanghai Fuxisi Technology Co. Ltd.

2014.7 – 2016.9

Software Development Engineer

- Built an e-commerce platform with Ruby on Rails 4 for compact smart-home devices (home gateways, IR remotes), covering product catalog, inventory, orders, payments, after-sales, and an admin dashboard.
- Designed and shipped RESTful mobile APIs to support app checkout and device management, including authentication, error model, pagination, and API versioning.
- Integrated payment and logistics callbacks to drive order state transitions; improved operator efficiency through configurable promotions and content management.

Augmentum, Inc.

2013.7 – 2014.6

Software Development Engineer (Java)

- Designed and implemented an internal ticketing system using Java, Spring MVC, Hibernate, and MySQL.
- Built workflows for ticket creation, assignment, SLA tracking, and role-based access control; provided RESTful APIs.
- Wrote unit and integration tests with JUnit and Mockito; participated in code reviews to improve maintainability.
- Used Maven and Jenkins for CI/CD and documented deployment runbooks for production releases.

Fastlink

2011.9 – 2012.6

R&D Engineer (Intern)

- Developed and maintained the cloud host administration dashboard covering VM provisioning, monitoring, and quota management.
- Maintained and improved the customer ticketing system to streamline incident handling with operations.
- Performed daily Linux server maintenance and monitoring; automated routine tasks with shell scripts.

Projects

RL from Human Feedback on a 32x NVIDIA A800 GPU Cluster *Core Member*

2023.7 – 2023.12

- Co-developed a **70B Reward Model** for reinforcement learning from human feedback (RLHF), based on **Llama2** and using a cluster of **16 NVIDIA A800** GPUs.
- Collaborated to build and organize the **training data pipeline**, achieving an improvement of approximately **14%–27%** over the 13B Reward Model on various internal evaluation metrics; provided **safety metrics** based on the reward model, aiding other safety-related projects.
- Led the development of **16B** and **1.9B critic/actor** models for RLHF, implementing **PPO** to reinforce the actor fine-tuning model; coordinated **multi-GPU** communication to implement **multi-model inference** across the **distributed** system.
- **Open Source Model:** github.com/bilibili/Index-1.9B

Distributed RL Systems Framework *Leader*

2020 – 2023

- Built a **ZeroMQ**-based messaging layer to stream episode trajectories, rewards, observations, and monitoring metrics across nodes.
- Decoupled services into **trainer**, **CPU/GPU inference**, **evaluator**, **monitor**, **model pool**, and a **data buffer** using **Arrow shared memory**.
- Added a **multi-agent** research service supporting **population-based methods** (**self-play**, **league training**, **PBT**) with real-time capability tracking (**Elo/TrueSkill**, win rate, cooperation metrics) and early-stop when regressions are detected.
- Designed an **RL-aware name server** to dynamically **rebalance actors** under cluster instability and to **recover failed workers**.
- Containerized the stack with **Docker** for reproducible multi-project deployments and rapid spin-up.
- Validated the framework across **Neural MMO**, **Google Research Football**, and multiple **Unity** titles, enabling code reuse and faster iteration.

Applied RL for Games and Live Interaction *Leader*

2020 – 2023

- Partnered with the game team to deploy **ML-Agents** in **Unity**; used **ray-based observations** in a **3D** world for coin collection, pathfinding, and obstacle avoidance.
- Trained agents headlessly by packaging the game into **no-GUI Docker** containers and distributing jobs on a **Linux cluster**.
- Iterated on **reward shaping** and **curriculum** to increase movement speed, scoring rate, and navigation stability.
- Launched **live interactive** demos where viewers issued text commands in the stream to control or co-play with agents.

Big Data Processing and Real-time Monitoring Framework

2016 – 2018

- Built and operated the offline data platform on **Hadoop/Hive**, standardizing **SQL ETL** and reporting jobs.
- Deployed **Apache Airflow**; designed **DAGs** for scheduling, dependencies, **SLAs**, alerting, retries, and backfills.
- Implemented a **streaming ETL** pipeline: **Kafka** producers from recommender services; **Go** consumers for real-time cleansing, filtering, and schema formatting.
- Cached processed streams in a distributed **Redis** cluster; persisted hourly to **HDFS**/object storage as partitioned datasets.
- Delivered near real-time experiment monitoring with **Ruby on Rails** and **Grafana**, tracking key metrics across concurrent **A/B tests** and surfacing anomalies.
- Provided curated datasets and recent-trend features for downstream **ML training** and **online learning**.
- Established basic **data governance**, including schema versioning and data-quality checks, to ensure reliable up-stream/downstream execution.

IFTTT-style Automation for Mobile & Smart Home

2014 – 2016

- Developed a Ruby-based IFTTT rule engine for phone and smart-home automation using a triggerconditionaction model; typical scenes included temperature/humidity thresholds to control air conditioners.
- Implemented both push and pull channels to accommodate heterogeneous devices and services.
- Exposed unified mobile APIs for creating, enabling, disabling, and monitoring automation tasks; added safeguards such as idempotency and basic retry to keep rules reliable.

Research Experience

Using MARL to Investigate Automated Vehicle Safety vs. Effectiveness Trade-Off

2024.1 – 2024.9

↔ Advisor: Prof. Peter Zhang

CMU

- Proposed **OPTIMA**, a distributed MARL framework for coordination-aware autonomous vehicles; modeled 40-agent unsignaled four-way intersections in MetaDrive with continuous control.
- Designed and compared **CTDE** vs. a centralized execution variant (**CTCE**) by pooling hidden states; quantified safetyefficiency trade-offs.
- Added rule-based reward shaping: *safe distance* and *right-of-way responsibility*; combined rules reduced crashes from 31.43 to 12.61 and out-of-road to 13.48 with minimal speed loss.
- Built a distributed training system on 4xV100 GPUs and 256 CPUs; separated inference/training, achieving ~15,000 samples/min and 256 updates/min; trained ~250M steps in 24h; PPO reached 100% success, outperforming SAC/DDPG.
- Mitigated async stale-data drift via a freshness constraint (avg policy-gap ≤ 8 versions) and controlled discards (~13%), improving stability without hurting throughput.
- **Open Source Code:** github.com/Passerby/right-of-way-rl

Neural MMO Multi-Agent Competitions *Leader*

2022.4 – 2022.10

- Built a **TensorFlow multi-agent PPO** system with a **central learner** and **distributed actors**; scaled training across **8 machines × 8 GPUs** and **32 machines × 64 CPU cores**.
- Engineered **opponent modeling** and **adaptive reward shaping**; ran scheduled ablations to tune weights for combat, exploration, and economy objectives.

- Optimized team role specialization and **inventory/combat** tactics for **NMMO v1.6**. Context: 8 agents per team on **128×128** maps versus 15 other teams for **1,024** ticks; finals executed **~10,000 matches**.
- Results: **1st place** (IJCAI 2022 PvP Final Top16) and **2nd place** (NeurIPS 2022 PvP Final Top16).
- Publications & talks: co-author of the **NeurIPS 2022 competition report**; invited strategy presentation at NeurIPS. Video: https://www.youtube.com/watch?v=nJqbIP_g4uM

Google Research Football Multi-Agent Competition *Leader*

2020.6 – 2021.4

- Built a **5v5** multi-agent control stack on **Google Research Football** using **parameter-sharing PPO** under **CTDE**, with **attention**-based role specialization to improve coordinated passing, pressing, and coverage.
- Designed a **league-style self-play** pipeline with **opponent sampling** and **curriculum**; automated evaluation on win rate, goal difference, and possession using **Elo/TrueSkill**-style ratings.
- Engineered a distributed rollout/learn/evaluate architecture (**Docker/K8s**) that decoupled **actors**, **learners**, and **model pool** for stable, high-throughput training.
- Results: **Runner-up**, IEEE CoG 2022 Football AI Competition Track 1; **Third place**, IEEE CoG 2023 Football AI Competition Track 1.

Research on the RL algorithms within the Dota2 environment *Core Member*

April 2019 – Aug 2021

- Developed a **large-scale distributed** RL framework for the **Dota 2 1v1** environment, utilizing **192 NVIDIA GTX 1080 Ti GPUs** and **1000 CPUs** for training and evaluation.
- Investigated multiple RL algorithms (**A2C**, **DQN**, **PPO**) under competitive **self-play** and **human-vs-AI** settings, establishing **evaluation metrics** to quantify agent improvement.
- Achieved a trained policy capable of defeating professional Dota 2 players with **MMR ~8000** in **1v1** match settings, demonstrating strong generalization and tactical adaptation.
- Designed and deployed **Docker**-based distributed pipelines, enabling reproducible large-scale training and efficient resource utilization.
- Implemented automated **feature extraction** from **undocumented Dota 2 APIs** to enable end-to-end state representation and policy optimization.
- **Open Source**: github.com/bilibili/LastOrder-Dota2

Publications

- Rui, Du, K. Zhao, J. Hou, Q. Zhang, and P. Zhang. Optima: Optimized policy for intelligent multi-agent systems enables coordination-aware autonomous vehicles. In *Proceedings of the 2025 IEEE Intelligent Transportation Systems Society Conference (ITSC)*. IEEE, 2025. URL <https://arxiv.org/submit/5909010/view>. Accepted.
- E. Liu, J. Suarez, C. You, B. Wu, B. Chen, J. Hu, J. Chen, X. Zhu, C. Zhu, J. Togelius, S. Mohanty, W. Hong, Rui Du, Y. Zhang, Q. Wang, X. Li, Z. Yuan, X. Li, Y. Huang, K. Zhang, H. Yang, S. Tang, and P. Isola. **The NeurIPS 2022 Neural MMO Challenge: A Massively Multiagent Competition with Specialization and Trade**, 2023. URL <https://arxiv.org/abs/2311.03707>.

Awards & Honors

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| 1 st place, IJCAI Conf | IJCAI 2022 the Neural MMO Challenge Final Result . | 2022 |
| 2 nd place, NeurIPS Conf | NeurIPS 2022 the Neural MMO Challenge Final Result . | 2022 |
| 2 nd place, IEEE CoG | IEEE CoG 2022 Google Football AI Challenge 1 Final Result . | 2022 |
| 3 rd place, IEEE CoG | IEEE CoG 2023 Google Football AI Challenge 1 Final Result . | 2023 |

Skills

Language proficiency	Mandarin (Native), English (Fluent)
Programming Languages	Python, Go, C++, Rust, Ruby, JavaScript, C#, SQL, Bash, L ^A T _E X
AI Algorithms & Areas	PPO, GRPO, SAC, DQN, A ₃ C, CTDE/CTCE, self-play, PBT
LLMs & Alignment	Reward Modeling; DPO, RLHF, RLVR;
Distributed Training & Inference	Megatron, Horovod; multi-GPU/multi-node; ZeroMQ, gRPC; Arrow shared memory; Docker, Kubernetes
Data Engineering	Hadoop, Hive, SQL ETL; Apache Airflow (DAGs, SLAs, alerts); Kafka
Monitoring & Experimentation	Grafana dashboards; near-real-time A/B testing; realtime feature pipelines
Simulation & Game Engines	Unity ML-Agents, Google Research Football, NMMO, MetaDrive, Dota 2
Evaluation & Metrics	Elo, TrueSkill, win rate, goal difference, success rate, crash/out-of-road rates
Tools & Frameworks	TensorFlow, PyTorch, JIT; Ruby on Rails; Git; Linux

Hobbies

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| Hobbies | Dota2 (MMR 5000), Video Games, Fitness |
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